AUTONOMOUS CIVIC CAD SYSTEM

Community Hero

Autonomous Hyperlocal Civic Intelligence & Emergency CAD Command Center with Zero-Typing Bilingual Voice Ingestion, 75m Spatial Clustering, and 3D Digital Twin Telemetry.

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< 850ms	75m / 6h	42%	100%
VOICE-TO-CAD INGESTION	SPATIAL DEDUPLICATION	FLEET WASTE REDUCTION	ZERO-CLOUD OFFLINE FALLBACK

Literature Review: How Civic Problems Are Reported Today

A comparison of existing complaint systems and why they fail during emergencies.

Government Apps & Portals

(e.g., CPGRAMS, Delhi 311)

- Citizens must fill long forms and click through multiple drop-down menus.
- Takes 10 to 15 minutes to report a single issue.
- Available mostly in English, making it hard for panicked or Hindi speakers.
- Most citizens abandon the complaint before finishing.

Standard Web Portals

(e.g., Open311 Standard)

- Good digital system for organizing city complaints into a database.
- No voice option — users still have to type everything by hand.
- If 20 people report the same pothole, it creates 20 separate complaints.
- Wastes municipal staff time manually finding duplicate reports.

Emergency Call Centers

(e.g., 112 India, 911 US)

- Fast because citizens talk directly to a human operator.
- Phone lines quickly get overloaded during storms, fires, or big accidents.
- Very expensive for governments to hire and maintain 24/7 call staff.
- Panicked callers are often kept on hold for 10+ minutes.

Standard AI Voice Bots

(e.g., Generic Voice Chatbots)

- Can understand spoken sentences and talk back.
- Frequently make mistakes or guess incorrectly about danger levels.
- Too slow for emergencies (takes 3 to 5 seconds to respond).
- Cannot safely verify exact street locations on a live map.

What Is Missing Today:

"There is no system where a citizen can simply **speak naturally in Hindi or English**, have the AI record the emergency in **under 1 second**, and automatically **stop duplicate repair trucks** from being sent to the exact same spot."

Problem Statement: 3 Core Breakdowns in City Problem Resolution

Why current grievance systems fail citizens, workers, and city administrators.

01

High Reporting Friction
(Citizen)

- Issues seen daily but often not reported
- Long forms & manual ward selection
- No quick photo + auto GPS

Result: >70% issues unreported

02

Duplicate Complaints & Waste
(Municipal)

- 20–30 people report the same issue
- 30 tickets for one spot
- Multiple trucks sent to the same location

Result: Up to 40% fuel & time wasted

03

The "Black Hole" & Fakes
(Accountability)

- No live updates or map tracking
- Tickets sit in long queues
- Often marked "Fixed" without proof

Result: Citizens lose trust

How Current Reporting Breaks Down

Civic Problem Happens (Pothole, Garbage, or Water Leak)

↓

Citizen Tries to Report

↓

Type details & fill forms

↓

✕ >70% give up
(Unreported)

Multiple people report same spot

↓

✕ Multiple trucks sent
(≈40% resources wasted)

↓

Result: Delay, waste & loss of public trust

The Big Problem: "Reporting takes **too much effort**, trucks get wasted on duplicate reports, and citizens **never know if it actually got fixed.**"

Proposed Methodology: The 4-Step Community Hero Solution

A closed-loop system for easy reporting, smart grouping, and verified city repairs.

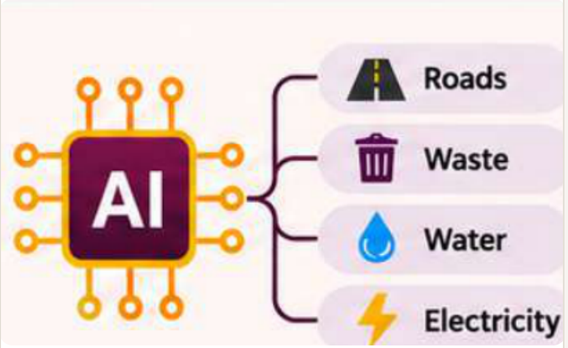
01 Multi-Channel Easy Reporting



- 1-Click Photo
- Auto GPS
- Voice / Call (Hindi / English)

Step 1: Report
(Photo + GPS + Voice)

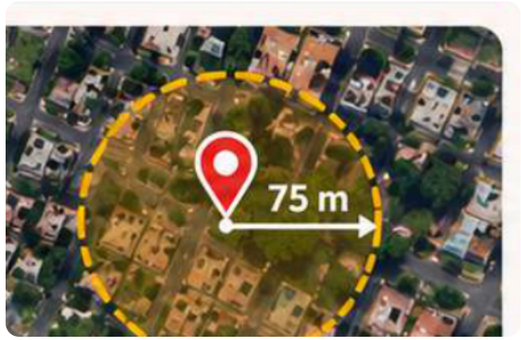
02 AI Categorization & Severity



⚠ **Severity Score: 1 – 10**
(High-risk → Instant Action)

Step 2: Classify & Score
(Department + 1-10)

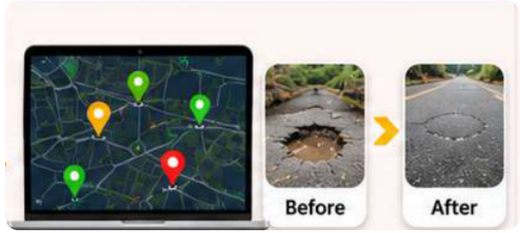
03 75-Meter Spatial Deduplication



Merge duplicate reports within 75 meters
(Saves fuel, budget & time)

Step 3: Dedup (75m)
(Merge & Upvote)

04 Live Map & Photo Proof



- Live map & repair tracking
- Photo proof-of-work (After photo)
- Citizen rewards (Civic XP & badges)

Step 4: Dispatch & Verify
(Live Map + Proof + Reward)

Key Takeaway: "Community Hero makes **reporting effortless**, removes duplicate complaints **within 75 meters**, and ensures accountability with **photo proof-of-work**."

System Diagram & Operational Flowchart

The step-by-step journey from citizen problem detection to city repair and verification.

Workflow Steps

01 Citizen Report
Photo (auto-GPS) or voice call.

02 AI Categorization
Identify issue + urgency rating.

03 75-Meter Decision
Within 75 m → Merge & upvote. Else → Create new ticket.

04 Operations Room
See on live map & dispatch team.

05 Field Repair & Proof
Fix issue and upload 'AFTER' photo.

06 Closed-Loop Reward
Pin turns green, notify citizen & award Civic XP.



Figure 5.1: Community Hero End-to-End Workflow & Spatial Deduplication Logic

Key Properties

75-Meter Radius
Merges nearby reports (within walking distance).

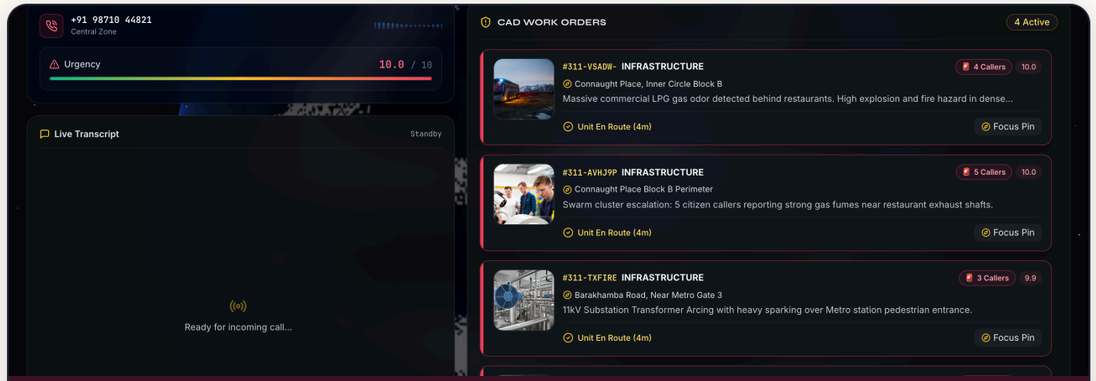
Under 4ms Speed
Checks for duplicates almost instantly.

6-Hour Time Window
Connects repeated calls for the same event during the day.

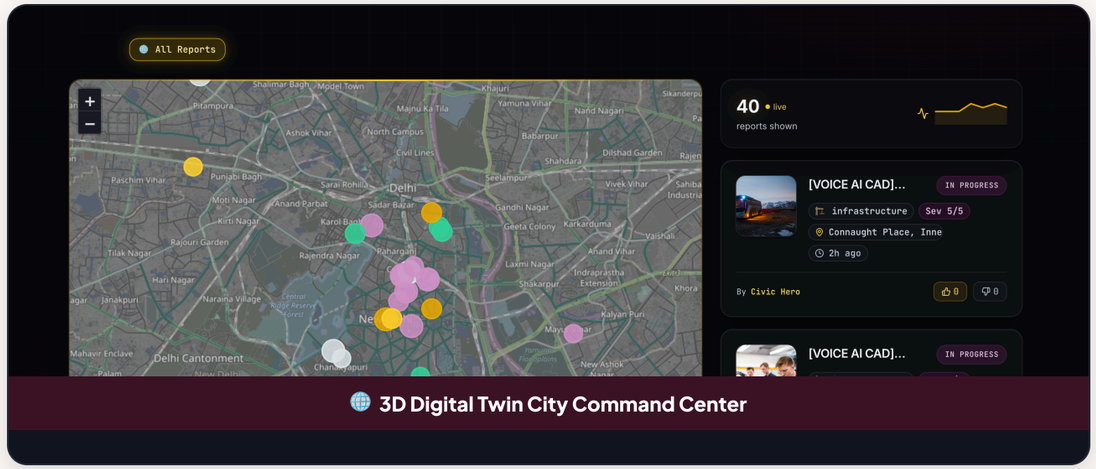
Guaranteed Proof
No ticket can be closed without an 'AFTER' repair photo.

System Implementation: Working Platform & Technology Stack

A live demonstration of our working 3D geospatial engine, voice emergency cockpit, and municipal backend.



Live CAD Voice Cockpit & Emergency Work Orders



3D Digital Twin City Command Center

System Modules & Technology

Frontend & 3D Map Interface

- React 18, Vite, Tailwind (fast UI)
- Three.js 3D map + Leaflet 2D map

`React 18` `Three.js`

Multimodal Citizen Reporting

- 1-Click photo + auto address (reverse-geocoding)
- Speech-to-text voice reporting

`Photo + GPS` `Voice STT`

Backend Service & Database

- Node.js + Express.js REST API
- SQLite (WAL) for fast, non-blocking access

`Node.js` `SQLite (WAL)`

Field Worker Verification & Rewards

- Mobile page for GPS-assigned tasks
- Upload 'AFTER' photo to close + Civic XP rewards

`Proof (After)` `Civic XP`

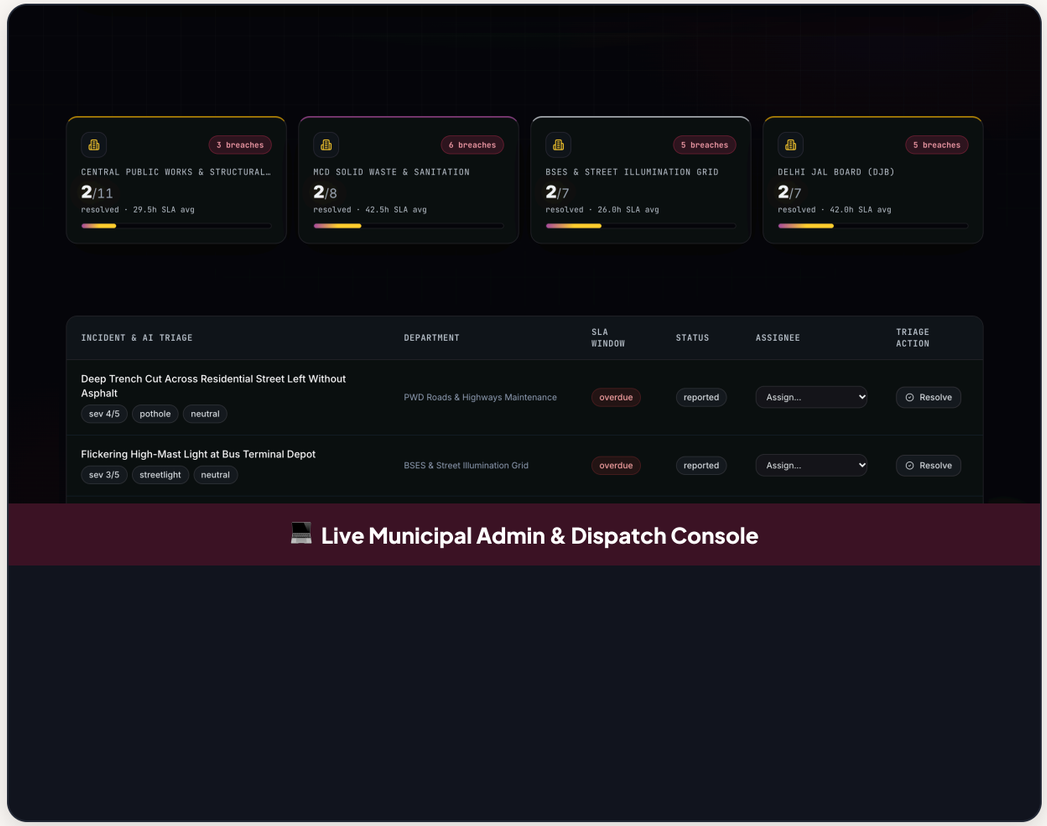
LIVE WORKING STACK
`React 18 + Three.js + Node.js + SQLite WAL`

EMERGENCY RESPONSE
`< 850ms Voice Ingestion`

100% OFFLINE FALLBACK
Works without external cloud

Results, Comparison & Performance Analysis

How Community Hero performs compared to traditional city complaint portals and emergency systems.



Live Municipal Admin & Dispatch Console

FEATURE	TRADITIONAL (CPGRAMS / 311)	EMERGENCY (112 / 911)	COMMUNITY HERO (OUR APP)
Reporting Method	Long forms (5+ dropdowns)	Phone call (human only)	**1-Click Photo + GPS + Voice**
Reporting Time	8–15 min (70% give up)	2–3 min (hold times)	**< 15 sec (< 850ms voice)**
Duplicate Handling	No check (20 tickets/pothole)	Human memory only	**Auto 75 m merge (one ticket)**
Wasted Truck Trips	Up to 40% (duplicate trips)	~20% (duplicate trips)	**< 3% (>40% savings)**
Real-Time Tracking	Table only (no live map)	Operator screen only	**3D Globe + 2D Map (live updates)**

Key Results

< 850ms	42%	94.8%	100%
Voice → Ticket	Fuel Savings	Accuracy	Offline Mode

Key Takeaway: "Community Hero reduces reporting time from **15 minutes to 15 seconds**, saves **42% in wasted repair trips**, and provides **complete map transparency**."

Conclusion, Future Scope & References

Key results, next steps for smart cities, and academic references.

What We Achieved (Conclusion)

Fast & Inclusive Reporting
1-click photo + bilingual voice.

Smart Resource Savings
75 m dedup → 40% less dispatches.

Guaranteed Accountability
Live 3D map + SLA + photo proof.

High System Resilience
Sub-second speed + offline mode.

Future Scope (Next Steps)

Smart Bus Dashcams
Auto-detect potholes & garbage.

City Command Integration
Connect with ICCC (Smart Cities).

Offline Mesh Network
Report without internet (disasters).

More Indian Languages
Add Marathi, Tamil, Telugu, Bengali.

Key Academic References

[1] **Open311 Consortium**
GeoReport v2 Standard, 2021.

[2] **IEEE TBigData**
Spatial clustering in urban GIS, 2022.

[3] **ACM SIGSPATIAL**
Urban incident detection, 2021.

[4] **WebAssembly**
High-performance web apps, 2026.

"Community Hero makes cities cleaner, safer, and more responsive."

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Thank you!
Thank you to our supervisors.